Permutation groups

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Preface

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Chapter 1

Two motivating examples

1.1 Finite groups of isometries

A $n \times n$ matrix A is *orthogonal* if $A^T A = I$. Since $\det(A^T A) = (\det A)^2$ the determinant of A is either $+1$ or -1. If A and B are orthogonal, then an easy calculation shows that AB^-1 is orthogonal as well. Therefore, $n \times n$ matrices form a subgroup of GL_n, called the *orthogonal group* O_n. Those elements of O_n which have determinant equal to $+1$ form a subgroup of O_n called the *special orthogonal group* SO_n.

It is easy to see that the linear mapping corresponding to an orthogonal matrix preserves distances in $\mathbb{R}^n$. Coversely, any linear mapping $\mathbb{R}^n \to \mathbb{R}^n$ that preserves lengths, preserves also distances and right angles, and thus maps the standard basis to an orthonormal basis. The matrix representing this mapping has the elements of this basis as its columns, so it is orthogonal.

Two and three dimensions. If $A \in \mathrm{O}_2$, the columns of A are unit vectors and are orthogonal to one another. Suppose that

$$A = \begin{pmatrix} a & c \\ b & d \end{pmatrix}.$$

Then (a, b) lies on the unit circle, giving $a = \cos\theta$, $b = \sin\theta$ for some θ satisfying $0 \le \theta < 2\pi$. Since (c, d) is orthogonal to (a, b) and also lies on the unit circle, we have $c = \cos\varphi$, $d = \sin\varphi$, where either $\varphi = \theta + \pi/2$ or $\varphi = \theta - \pi/2$. We obtain

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \quad \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}.$$

The first matrix is an element of SO_2 and represents the anticlockwise rotation through θ. The second has determinant -1 and represents the reflection in a line at angle $\theta/2$ to the positive x-axis.

Now suppose that $A \in \mathrm{SO}_3$. The characteristic polynomial $\det(A - \lambda I)$ is a cubic and therefore must have at least one real root, i.e., A has a real eignevalue. Since the determinant is the product of eigenvalues, it follows that $+1$ is an eigenvalue of A. If $\mathbf{v}$ is a corresponding eigenvector, the line through the origin determined by $\mathbf{v}$ is left fixed by A. Also since A preserves right angles, it must map the plane which is perpendicular to $\mathbf{v}$, and which contains the origin, to itself. We can construct an orthonormal basis for $\mathbb{R}^3$ with the unit vector $\mathbf{v}/\|\mathbf{v}\|$ as the first element. The matrix with respect to this new basis corresponding to same mapping as A will be an element of SO_3 taking the form

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & b_{11} & b_{12} \\ 0 & b_{21} & b_{22} \end{pmatrix}, \quad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \in \mathrm{SO}_2,$$

So A is a rotation with axis determined by $\mathbf{v}$. Conversely every rotation of $\mathbb{R}^3$ which fixes the origin is represented by a matrix in SO_3.

If $A \in \mathrm{O}_3$, but $A \notin \mathrm{SO}_3$, then $AU \in \mathrm{SO}_3$, where

$$U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

The matrix U represents a reflection in (x, y)-plane. Clearly,

$$A = (AU)U$$

and since AU is a rotation, A is a reflection in the (x, y)-plane followed by a rotation.

Proposition 1.1. $\mathrm{O}_3 \cong \mathrm{SO}_3 \times \mathbb{Z}_2$.

Proof. Let I denote the 3×3 identity matrix. Both I and $-I$ commute with every other matrix in O_3, and together they form a subgroup of O_3 of order 2 isomorphic to $\mathbb{Z}_2$. Define

$$\varphi\colon \mathrm{SO}_3 \times \{I, -I\} \to \mathrm{O}_3, \quad \text{by} \quad (A, U) \longmapsto AU,$$

where $A \in \mathrm{SO}_3$ and $U \in \{I, -I\}$.

We have

$$\varphi((A, U)(B, V)) = \varphi(AB, UV) = ABUV = AUBV = \varphi(A, U)\varphi(B, V),$$

for all $A, B \in \mathrm{SO}_3$ and $U, V \in \{I, -I\}$, so φ is a homomorphism.

If $\varphi(A, U) = \varphi(B, V)$, then $AU = BV$, giving $\det(AU) = \det(BV)$. But

$$\det(AU) = \det(A)\det(U) = \det(U)$$

because $A \in \mathrm{SO}_3$, and similarly $\det(BV) = \det(V)$. Hence $U = V$, $A = B$, and we conclude that φ is injective. It only remains to check that φ is surjective. Given $A \in \mathrm{O}_3$, either $A \in \mathrm{SO}_3$, in which case $A = \varphi(A, I)$, or $A(-I) \in \mathrm{SO}_3$ and $A = \varphi(A(-I), -I)$. Thus, φ is an isomorphism. □

Remark 1.2. The same argument shows that $\mathrm{O}_n \cong \mathrm{SO}_n \times \mathbb{Z}_2$ only *when n is odd*. The map φ is not an isomorphism for an even n since then we have $\operatorname{Ker}\varphi = \{(I, I), (-I, -I)\}$. In fact, for even $n \geq 4$ the statement is not true: for instance, the groups O_n and $\mathrm{SO}_n \times \mathbb{Z}_2$ have centers of different orders, two and four, respectively. And for $n = 2$, the group SO_2 is abelian, so $\mathrm{SO}_2 \times \mathbb{Z}_2$ has much larger center than O_2.

Theorem 1.3. *A finite subgroup of* O_2 *is either cyclic or dihedral.*

Proof. Let G be a finite non-trivial subgroup of O_2. First, suppose that G lies inside SO_2 so that each element of G represents a rotation of the plane. Write A_θ for the matrix which represents the anticlockwise rotation through θ about the origin, where $0 \leq \theta < 2\pi$, and choose $A_\varphi \in G$ so that φ is positive and as small as possible.

Given $A_\theta \in G$, divide θ by φ to produce

$$\theta = k\varphi + \psi$$

where $k \in \mathbb{Z}$ and $0 \leq \psi < \varphi$. Then

$$A_\theta = A_{k\varphi+\psi} = (A_\varphi)^k A_\psi \qquad \text{and} \qquad A_\psi = (A_\varphi)^{-k} A_\theta.$$

Since A_θ and A_φ both lie in G, we see that A_ψ is also in G. This gives $\psi = 0$, since otherwise we contradict our choice of φ. Therefore, G is generated by A_φ and is cyclic.

If G is not contained inside SO_2, we set $H = G \cap \mathrm{SO}_2$. Then H is a subgroup of G which has index 2, and by the first part H is cyclic because it is contained in SO_2. Choose a generator A for H and an element B from $G \setminus H$. As B represents a reflection we have $B^2 = I$. If $A = I$, then G consists of I and B and is a cyclic group of order 2. Otherwise, the order of A is an integer $n \geq 2$. The elements of G are now

$$I,\ A,\ \ldots,\ A^{n-1},\ B,\ AB,\ \ldots,\ A^{n-1}B$$

and they satisfy $A^n = I$, $B^2 = I$, $BA = A^{-1}B$. The presentation

$$\langle A, B \mid A^n = I, B^2 = I, BA = A^{-1}B \rangle$$

determines the dihedral group $\mathbb{D}_n$. □

Theorem 1.4. *A finite subgroup of* SO_3 *is isomorphic either to a cyclic group, a dihedral group, or the rotational symmetry group of one of the Platonic solids.*

Proof. Let G be a finite subgroup of SO_3. Each element of G, other than the identity, represents a rotation of $\mathbb{R}^3$ about an axis which passes through the origin. We will work with rotations rather than the corresponding matrices. The two points where the axis of a rotation $g \in G$ meets the unit sphere are called the *poles* of g. These poles are the only points on the unit sphere which are fixed by the given rotation.

Let X denote the set of all poles of all elements of $G \setminus \{e\}$. Suppose $x \in X$ and $g \in G$. Let x be a pole of the element $h \in G$. Then

$$(ghg^{-1})(g(x)) = g(h(x)) = g(x),$$

which shows that $g(x)$ is a pole of ghg^{-1} and hence $g(x) \in X$. Therefore, we have an action of G on X. The idea of the proof is to apply the orbit-counting lemma to this action and show that X has to be a particularly nice configuration of points.

Let N denote the number of distinct orbits, choose a pole from each orbit, and call these poles $x_1, x_2, \ldots, x_N$. Every element of $G \setminus \{e\}$ fixes precisely two poles, while the identity fixes them all, so the orbit-counting lemma gives

$$N = \frac{1}{|G|}(2(|G|-1) + |X|) = \frac{1}{|G|}\left(2(|G|-1) + \sum_{i=1}^{N} |x_i^G|\right).$$

This rearranges to

$$\begin{aligned}
2\left(1-\frac{1}{|G|}\right) &= N - \frac{1}{|G|}\sum_{i=1}^{N} |x_i^G| \\
&= N - \sum_{i=1}^{N} \frac{1}{|G_{x_i}|} \\
&= \sum_{i=1}^{N}\left(1-\frac{1}{|G_{x_i}|}\right).
\end{aligned}$$

Assuming G is not the trivial group, the left-hand side of the above expression is greater than or equal to 1 and less than 2. But each stabilizer G_x has order at least 2 so that

$$\frac{1}{2} \leq 1 - \frac{1}{|G_{x_i}|} < 1,$$

for $1 \leq i \leq N$. Therefore, N is either 2 or 3.

If $N = 2$, the above equations give

$$2 = |x_1^G| + |x_2^G|$$

and there can only be two poles. These poles determine an axis L and every element of $G \setminus \{e\}$ must be a rotation about this axis. The plane which passes through the origin and which is perpendicular to L is rotated on itself by G. Therefore, G is isomorphic to a subgroup of SO_2 and has to be cyclic by Theorem 1.3.

If $N = 3$, writing x, y, z instead of x_1, x_2, x_3, we have

$$2\left(1-\frac{1}{|G|}\right) = 3 - \left(\frac{1}{|G_x|} + \frac{1}{|G_y|} + \frac{1}{|G_z|}\right)$$

and, therefore,

$$1 + \frac{2}{|G|} = \frac{1}{|G_x|} + \frac{1}{|G_y|} + \frac{1}{|G_z|}.$$

The sum of the three terms on the right-hand side is greater than 1, so there are only four possibilities for $(|G_x|, |G_y|, |G_z|)$:

$$(2,2,n), \quad \text{for any } n \geq 2, \quad (2,3,3), \quad (2,3,4), \quad (2,3,5).$$

From this it is possible to deduce the theorem. The theorem follows by carefully analyzing these case, we omit this part here. □

Remark 1.5. The rotational symmetry group of the tetrahedron is isomorphic to $\mathbb{A}_4$. Cube and octahedron have rotational symmetry group isomorphic to $\mathbb{S}_4$. Dodecahedron and icosahedron have rotational symmetry group isomorphic to $\mathbb{A}_5$.

Corollary 1.6. *If G is a finite subgroup of O_3, then G is isomorphic to a subgroup of one the following groups: $\mathbb{Z}_n \times \mathbb{Z}_2$, $n \geq 1$, $\mathbb{D}_n \times \mathbb{Z}_2$, $n \geq 2$, $\mathbb{A}_4 \times \mathbb{Z}_2$, $\mathbb{S}_4 \times \mathbb{Z}_2$, $\mathbb{A}_5 \times \mathbb{Z}_2$.*

1.2 Automorphism groups of graphs

Recall that a *(simple) graph* is a tuple $X = (V, E)$ such that $E \subseteq \binom{V}{2}$. An *automorphism* of X is a bijection $f\colon V \to V$ such that $\{x, y\} \in E \iff \{f(x), f(y)\} \in E$. The group of all automorphisms of X is denoted by $\mathrm{Aut}(X)$.

Theorem 1.7 (Frucht's theorem)**.** *For every finite group G, there is a graph X such that $G \cong \mathrm{Aut}(X)$.*

It is reasonable to consider what finite groups can be realized as automorphism groups of some restricted class of graphs. For instance, we can consider the class $\mathcal{T}$ of finite groups that can be realized as automorphism groups of trees.

Disconnected graphs. In order to characterize the groups belonging to $\mathcal{T}$, we will first derive a useful formula. Suppose that we have a disconnected graph X. We would like to be able to express its automorphism group in terms of its connected components. If X is a disjoint union of two non-isomorphic connected components Y and Y', then clearly $\mathrm{Aut}(X) \cong \mathrm{Aut}(Y) \times \mathrm{Aut}(Y')$. However, if X consists of, say, $k \geq 2$ copies of the same graph Y, then $\mathrm{Aut}(X)$ can no longer be expressed as a direct product.

In order to deal with this situation, we use the wreath product. Suppose that G acts on the set Ω. For each $\omega \in \Omega$, take a copy H_ω of a group H. The *wreath product* $H \wr G$ is the semidirect product of

$$K = \prod_{\omega \in \Omega} H_\omega$$

by G, where the homomorphism $G \to \mathrm{Aut}(K)$, required for the semidirect product, is defined naturally by the action of G on the coordinates of K.

As an example, we can apply this to the situation described above. In particular, we get the following

$$\mathrm{Aut}(X) \cong \mathrm{Aut}(Y) \wr \mathbb{S}_k.$$

In general, we have the following formula.

Theorem 1.8. *Let $X_1, \ldots, X_n$ be pairwise non-isomorphic graphs and let X be the disconnected graph consisting of $k_i \in \mathbb{N}$ copies of the graph X_i, for $i = 1, \ldots, n$. Then*

$$\mathrm{Aut}(X) \cong \prod_{i=1}^{n} \mathrm{Aut}(X_i) \wr \mathbb{S}_{k_i}.$$

Using Theorem 1.8, one can easily determine the automorphism group, provided that the automorphism groups of the connected components are known. We note that this can be further generalized to 2-connected and 3-connected components.

Trees. We are ready to characterize automorphism groups of trees. First, we reduce this problem to *rooted trees*, which are just trees with a distinguished vertex that has to be preserved by every automorphism.

Let T be a tree. The *center* of a tree is a set of vertices v that minimize the quantity

$$\max_{u \in V(T)} d(v, u),$$

where $d(v, u)$ is the length of the shortest path from v to u in T. Clearly, such vertices lie on every path of maximum length in T, and the center consists of either one or two vertices, based on the parity of this length. Moreover, the center is preserved by every automorphism. If the center consists of one vertex, we can make it a root. If it is of size two, we subdivide the edge between the two vertices by a new vertex, which will be the new root. Thus, we obtain a new rooted tree T_r with a root r such that $\mathrm{Aut}(T_r) \cong \mathrm{Aut}(T)$. On the other hand, if we start with a rooted tree T_r, we can attach a very long path to r that will force the center to be on this path. This gives an unrooted tree T with $\mathrm{Aut}(T) \cong \mathrm{Aut}(T_r)$.

Theorem 1.9 (Jordan, 1869)**.** *The class $\mathcal{T}$ of automorphism groups of trees can be inductively described as follows:*

(1) $\{1\} \in \mathcal{T}$.
(2) If $G, H \in \mathcal{T}$, *then* $G \times H \in \mathcal{T}$.
(3) If $G \in \mathcal{T}$, *then* $G \wr \mathbb{S}_n \in \mathcal{T}$, *for* $n \geq 2$.

Proof. By the argument above, it is sufficient consider only rooted trees. First we need to argue that all the operations (1)–(3) can be realized by some tree, which we do by induction.

Clearly the trivial group is the automorphism group of a tree with one vertex If $G, H \in \mathcal{T}$, then by induction hypothesis there are trees T_G and T_H that realize G and H, respectively. If G and H are not isomorphic then we can construct a tree T with $\mathrm{Aut}(T) \cong G \times H$ by attaching the roots of T_G and T_H to a common new root. If $G \cong H$, this construction creates new automorphisms. We fix this by subdividing one of the newly created edges. Finally, if $G \in \mathcal{T}$ and T_G realizes G, then we can construct T with $\mathrm{Aut}(T) \cong G \wr \mathbb{S}_n$ by attaching the roots of n copies of T_G to a common new root. We proved that every group in the class $\mathcal{T}$ is in fact an automorphism group of some tree.

It remains to show that for a rooted tree T, its automorphism group is isomorphic to a group in $\mathcal{T}$. We again proceed by induction. If T has one vertex, then the statement clearly holds. If T has more vertices, then we remove the root and get a forest of rooted trees. The automorphism group is determined by the formula in Theorem 1.8. It is clear that the only operations that appear in the formula are the direct product and the wreath product with symmetric groups. $\square$

Chapter 2

Basic notions

We deal only with finite groups; therefore, throughout the following, all groups are finite, and the sets on which they act are finite as well.

2.1 Actions of groups on sets

Definition 2.1. Let G be a group and Ω a set. A function

$$\omega\colon \Omega \times G \to \Omega, \qquad (\alpha, g) \mapsto (\alpha, g)\omega$$

is called an *action of G on Ω* if the following conditions hold:

1. $(\alpha, 1_G)\omega = \alpha$ for all $\alpha \in \Omega$.
2. $(\alpha, gh)\omega = ((\alpha, g)\omega, h)\omega$ for all $\alpha \in \Omega$ and $g, h \in G$.

The set Ω is called a *G-set*, and we say that *G acts on Ω*. Furthermore, $|\Omega|$ is called the *degree* of the action of G on Ω.

Remark 2.2. In general, we use the notation preferred by Wielandt, that is, we write α^g for $(\alpha, g)\omega$. Then the two conditions above can be reformulated as

1. $\alpha^{1_G} = \alpha$ for all $\alpha \in \Omega$.
2. $\alpha^{gh} = (\alpha^g)^h$ for all $\alpha \in \Omega$ and $g, h \in G$.

Definition 2.3. Let G be a group acting on a set Ω. If $\alpha^g = \alpha$ for some $\alpha \in \Omega$ and $g \in G$, then we say that *g fixes the element α*. The set

$$G_{(\Omega)} = \{g \in G \mid \alpha^g = \alpha \text{ for all } \alpha \in \Omega\}$$

is called the *kernel* of the action of G on Ω. If $G_{(\Omega)} = \{1_G\}$, then the action is called *faithful*.

Definition 2.4. Let G be a group and Ω a set. A homomorphism

$$\varphi\colon G \to \mathrm{Sym}(\Omega)$$

is called a *permutation representation of G on Ω*.

Theorem 2.5. *Let G be a group and Ω a set.*

1. *Given an action* $\omega\colon \Omega \times G \to \Omega$ *of* G *on* Ω*, there exists a permutation representation* φ *of* G *on* Ω *defined by*

$$g\varphi = \varphi_g, \qquad \textit{where } \varphi_g \in \mathrm{Sym}(\Omega) \textit{ and } \alpha\varphi_g = (\alpha, g)\omega.$$

2. *Conversely, a permutation representation* φ *of* G *on* Ω *defines an action* $\omega\colon \Omega \times G \to \Omega$ *of* G *on* Ω *by*

$$(\alpha, g)\omega = \alpha^{g\varphi}.$$

Thus, it is not necessary to distinguish between actions of a group G on a set Ω and permutation representations of G on Ω. If G acts faithfully on Ω, then G is isomorphic to $(G)\varphi$.

Example 2.6.

(1) The symmetric group $\mathrm{Sym}(\Omega)$ acts on Ω via the natural action α^g, where $\alpha^g = (\alpha)g$, the image of $\alpha \in \Omega$ under g.

(2) A group acts on itself by right multiplication. Let G be a group and let $\Omega = G$. Define

$$\rho\colon \Omega \times G \to \Omega, \qquad (a, g) \mapsto (a, g)\rho = ag.$$

Then ρ is an action, since $(a, 1_G)\rho = a1_G = a$ for all $a \in \Omega = G$, and

$$(a, gh)\rho = agh = ((a, g)\rho, h)\rho$$

for all $a \in \Omega = G$ and $g, h \in G$.
The kernel $G_{(\Omega)}$ of this action is

$$G_{(\Omega)} = \{g \in G \mid ag = a \ \forall a \in \Omega\} = \{g \in G \mid hg = h \ \forall h \in G\} = \{1_G\}.$$

This action is called the *right regular action* of G on Ω.

(3) A group acts on itself by left multiplication. Let G be a group and let $\Omega = G$. Define

$$\omega\colon \Omega \times G \to \Omega, \qquad (a, g) \mapsto (a, g)\omega = g^{-1}a.$$

Then ω is an action, since $(a, 1_G)\omega = 1_G^{-1}a = a$ for all $a \in \Omega = G$, and

$$(a, gh)\omega = (gh)^{-1}a = h^{-1}g^{-1}a = ((a, g)\omega, h)\omega$$

for all $a \in \Omega = G$ and $g, h \in G$.
The kernel $G_{(\Omega)}$ of this action is $G_{(\Omega)} = \{1_G\}$.
This action is called the *left regular action* of G on Ω.

(4) Let Ω be a finite set with $|\Omega| = n$ and let $k < n$. Denote by $\binom{\Omega}{k}$ the set of all k-subsets of Ω. If G acts faithfully on Ω, then G also acts on $\binom{\Omega}{k}$, where for $A \in \binom{\Omega}{k}$ the action of $g \in G$ on A is defined by

$$A^g = \{a^g \mid a \in A\}.$$

The kernel $G_{(\Omega)}$ of this action is

$$\begin{aligned} G_{(\Omega)} &= \{g \in G \mid \alpha^g = \alpha \ \forall \alpha \in \Omega\} \\ &= \{g \in G \mid A^g = A \ \forall A \in \tbinom{\Omega}{k}\} \\ &= \{1_G\}. \end{aligned}$$

(5) Let G be a group and $\Omega = G$. Then G acts on Ω by

$$h^g = g^{-1}hg.$$

This action is called the *action of G on itself by conjugation.*
The kernel of this action is

$$\begin{aligned} G_{(\Omega)} &= \{g \in G \mid h^g = h \ \forall h \in \Omega\} \\ &= \{g \in G \mid hg = gh \ \forall h \in G\} \\ &= Z(G). \end{aligned}$$

(6) Let G be a group and Ω the set of all subgroups of G. Then G acts on Ω by

$$H^g = \{h^g \mid h \in H\}.$$

This action is called the *action of G on its subgroups by conjugation.*

(7) Let G be a group and H a subgroup of G. Then G acts on the set Ω of right cosets $\mathcal{R}_{G:H}$ of H in G by right multiplication via

$$(Hx)^g = Hxg.$$

This action is called the *action of G on the right cosets of H.*
The kernel of this action is

$$\begin{aligned} G_{(\Omega)} &= \{g \in G \mid (Hx)^g = Hx \ \forall Hx \in \Omega\} \\ &= \{g \in G \mid Hxg = Hx \ \forall Hx \in \Omega\} \\ &= \{g \in G \mid g \in x^{-1}Hx \ \forall Hx \in \Omega\} \\ &= \bigcap_{x \in G} x^{-1}Hx. \end{aligned}$$

In general, this action is not faithful. The group $G_{(\Omega)}$ is called the *G-core* of H. It is the largest normal subgroup of G contained in H.

(8) Let G be a subgroup of $\mathrm{GL}(n, \mathbb{F})$, the group of invertible $n \times n$ matrices over a field $\mathbb{F}$. Let $V = \mathbb{F}^n$. Then G acts on V by

$$v^g = vg.$$

The kernel of this action is

$$G_{(\Omega)} = \{g \in G \mid vg = v \ \forall v \in V\} = \{I\}.$$

(9) Let G be a subgroup of $\mathrm{GL}(n, \mathbb{F})$ and let $\mathrm{PG}(V)$ be the set of all 1-dimensional subspaces of V. Then G acts on $\mathrm{PG}(V)$ by

$$\langle v \rangle^g = \langle vg \rangle.$$

The kernel of this action is

$$\begin{aligned} G_{(\Omega)} &= \{g \in G \mid \langle vg \rangle = \langle v \rangle \ \forall \langle v \rangle \in PG(V)\} \\ &= \{aI \mid a \in \mathbb{F}\}. \end{aligned}$$

The group $G/G_{(\Omega)}$ is called the *projective general linear group.*

2.2 Similar actions

If a group G acts on a set Ω, then for $g \in G$ we denote by $\bar{g}_\Omega \colon \Omega \to \Omega$, $\alpha \mapsto \alpha^g$, the map induced by g.

Definition 2.7 (Equivariance)**.** Let the group G act on the sets Ω and Γ. A map $\lambda \colon \Omega \to \Gamma$ is called *G-compatible* (or *equivariant*) if

$$(\alpha^g)\lambda = (\alpha)\lambda^g \qquad \text{for all } \alpha \in \Omega,\ g \in G.$$

This means that the diagram

$$\begin{tikzcd} \Omega \arrow[r, "\bar{g}_\Omega"] \arrow[d, "\lambda"'] & \Omega \arrow[d, "\lambda"] \\ \Gamma \arrow[r, "\bar{g}_\Gamma"] & \Gamma \end{tikzcd}$$

is commutative for all $g \in G$. If, in addition, λ is bijective, then the actions (or the G-sets) are called *G-similar*, and λ is then called a *G-similarity*.

Let G be a group acting on Ω and let H be a group acting on Γ. Then G and H are called *permutation isomorphic* if there exists a bijection $\lambda \colon \Omega \to \Gamma$ and an isomorphism $\varphi \colon G \to H$ such that

$$(\alpha^g)\lambda = (\alpha)\lambda^{(g)\varphi} \qquad \text{for all } \alpha \in \Omega,\ g \in G.$$

Definition 2.8. Let G be a group acting on Ω. The action is called *transitive* if for every pair $(\alpha, \beta) \in \Omega \times \Omega$ there exists $g \in G$ such that $\alpha^g = \beta$.

Theorem 2.9 (Fundamental theorem for transitive G-sets)**.** *Let G be a group.*

(1) Let $H \leq G$. Then G acts transitively on the set $\mathcal{R}_{G:H}$ of right cosets by right multiplication:

$$\mathcal{R}_{G:H} \times G \to \mathcal{R}_{G:H}, \qquad (Ha, g) \mapsto Hag.$$

(2) Let G act transitively on the set Ω, let $\alpha \in \Omega$ and let $G_\alpha := \mathrm{Stab}_G(\alpha)$. Then Ω and $\mathcal{R}_{G:G_\alpha}$ are G-similar as G-sets:

$$\lambda \colon \Omega \to \mathcal{R}_{G:G_\alpha}, \qquad \alpha^g \mapsto G_\alpha g$$

is a G-similarity.

(3) Every transitive G-set Ω determines a conjugacy class $\mathcal{U}_\Omega$ of subgroups of G, namely

$$\mathcal{U}_\Omega := \{G_\alpha \mid \alpha \in \Omega\},$$

and the map

$$\Omega \to \mathcal{U}_\Omega, \qquad \alpha \mapsto G_\alpha$$

is G-equivariant. Moreover, Ω is G-similar to a transitive G-set Γ if and only if $\mathcal{U}_\Omega = \mathcal{U}_\Gamma$.